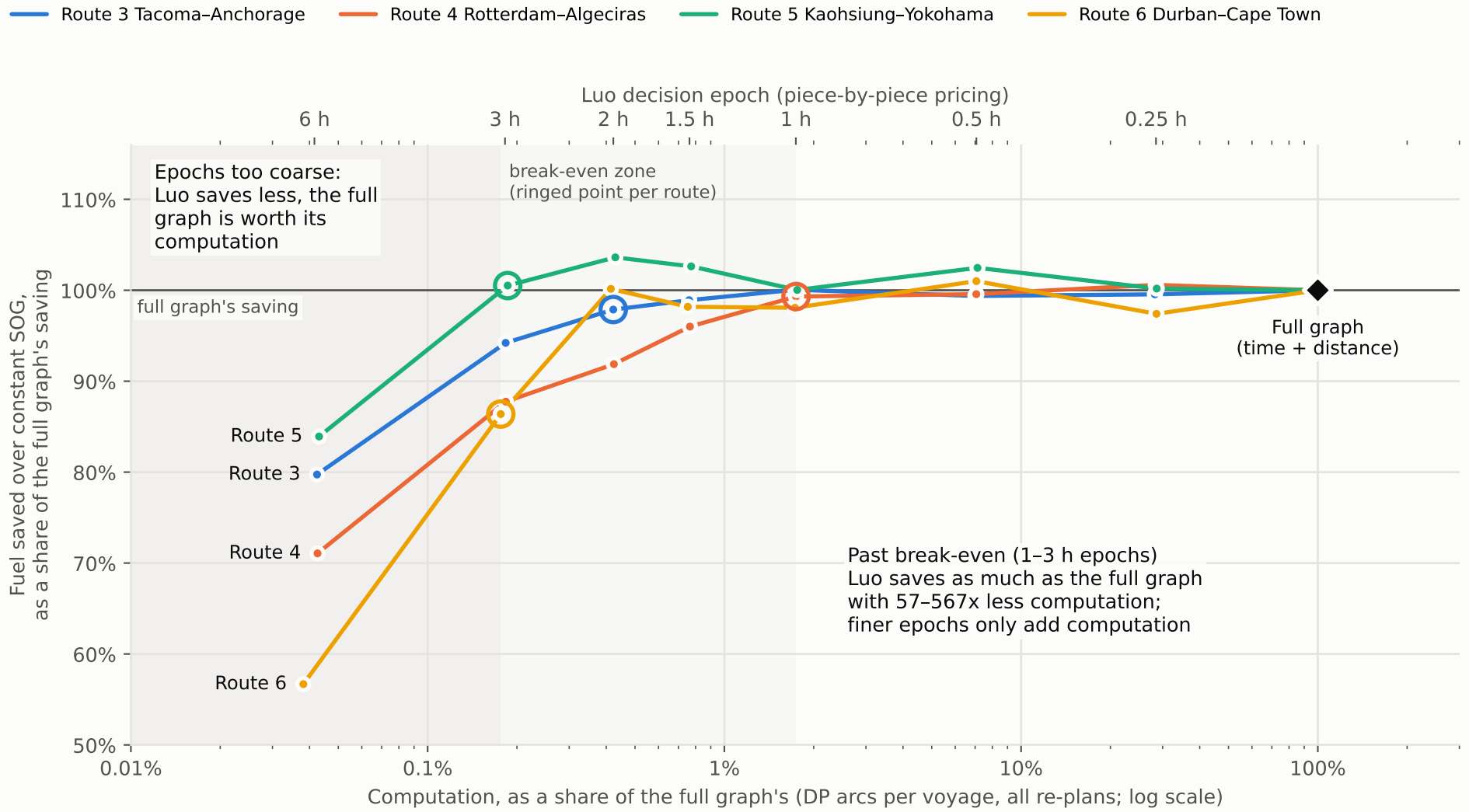
Below 1–3 h epochs the full graph saves more; from there on, Luo's time-only graph matches it far more cheaply



Routes 3–6, 25 voyages (departures every 48 h, 21 Oct – 2 Nov 2026), rolling horizon re-planned every 6 h with the forecast known 6 h earlier; route means. Not shown: Luo et al. as published (6 h, start-point pricing) saves -0.7% to -3.1% vs constant SOG, i.e. it burns more.